Mechanisms · Staff Answers

Learning goal: I can identify levers, pulleys, and gears and explain how they let a small effort move a heavy load.

Arrival Answers

1. Box 2 uses a lever and pivot.
2. The load.
3. The effort.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. Moving the effort end through a longer distance multiplies the force at the load end.

Hinge: A. Meshed gears turn in opposite directions; driving a larger gear reduces speed and increases turning force.

Evidence Desk Cards

Crowbar lifting a heavy paving slab: Lever: a long effort arm around a pivot multiplies the lifting force.

Window blind cord or flagpole rope: Pulley: pulling down on the rope lifts the blind or flag up.

Bicycle rear cassette and front chainring: Gears: changing gear teeth ratios changes pedalling force and speed.

Wheelbarrow carrying soil: Lever: the wheel axle acts

as the pivot while the handles provide a long effort arm.

Two-wheel workshop hoist: Pulley system: shares the load across rope sections so a smaller pull lifts the engine.

A lever gives free energy with zero movement: False claim: to use a smaller effort force, you must push through a longer distance.

Check & Repair Answer

Mechanisms trade distance for force: a long push with a small force lifts a heavy load a short distance.

Exit Ticket Answers

1. The pivot (or fulcrum). 2. Close to the load and far from the effort. 3. They apply a smaller effort force over a longer distance (or more turns) to produce a larger force on the load over a shorter distance.